

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 08/01/2026**
Time: 3 Hrs **GTM - 12** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. **Assertion A :** If Oxygen ion $\left(O^{-2}\right)$ and Hydrogen ion $\left(H^{+}\right)$ enter normal to the magnetic field with equal momentum, then the path of O^{-2} ion has a smaller curvature than that of H^{+}.

Reason R: A proton with same linear momentum as an electron will form a path of smaller radius of curvature on entering a uniform magnetic field perpendicularly.

In the light of the above statements, choose the ***correct*** answer from the options given below

1) Both **A** and **R** are true and **R** is the correct explanation of **A**

2) **A** is false but **R** is true

3) **A** is true but **R** is false

4) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**

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2. A particle starts moving with initial velocity from the origin of co-ordinates at time t = 0 and moves in the xy plane with a constant acceleration α in the y-direction. Its equation of motion is $y = \beta x^2$.

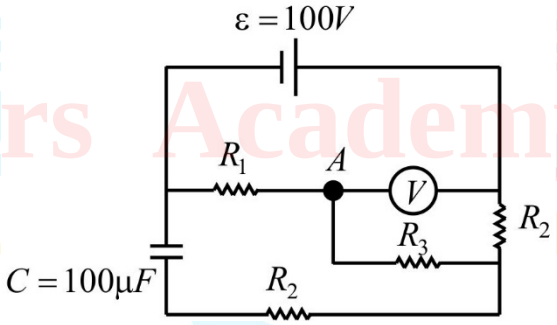
Its velocity component in the x-direction is:

1) α / β 2) $\sqrt{2\alpha / \beta}$ 3) $\alpha / 2\beta$ 4) $\sqrt{\alpha / 2\beta}$

3. In a two slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by 5×10^{-2} m towards the slits, the change in fringe width is $3\times10^{-5}\,m$. If the distance between the slits is 10^{-3} m, calculate the wavelength of the light used.

1) 4500 Å 2) 3000 Å 3) 9000 Å 4) 6000 Å

4. In the circuit shown in figure E is a battery of emf 100 V. Resistance $R_1 = 100\Omega, R_2 = 200\Omega, R_3 = 200\Omega$. The voltmeter resistance is 200Ω. What reading will the voltmeter show in steady state?



1) 50/3 V 2) 400/7 V 3) 100/3 V 4) 50 V

5. A solid steel ball of diameter 3.6 mm acquired terminal velocity $2.45\times10^{-2}\,m/s$ while falling under

Gravity through an oil of density $925\,kg\,m^{-3}$. Take density of steel as $7825\,kg\,m^{-3}$ and g as $9.8\,m/s^2$

The viscosity of the oil in SI unit is

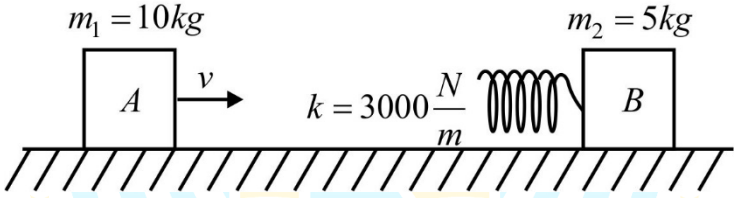
1) 1.99 2) 1.68 3) 2.18 4) 2.38

6. A resistance R draws P power when connected to an AC source. If an inductance is now placed in series with the resistance, such that the impedance of the circuit becomes Z, the power drawn will be

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1) $P\left(\frac{R}{Z}\right)^2$ 2) $P\left(\frac{R}{Z}\right)$ 3) $P\sqrt{\frac{R}{Z}}$ 4) P

7. Consider two blocks A and B of masses $m_1 = 10\ kg$ and $m_2 = 5\ kg$ that are placed on a frictionless table. The block A moves with a speed $v = 3m/s$ towards the block B kept at rest. A spring with spring constant $k = 3000\ N/m$ is attached with the block B as shown in the figure. After the collision, suppose that the blocks A and B, along with the spring in constant compression state, move together, then the compression in the spring is, (Neglect the mass of the spring)



1) 0.1m 2) 0.3m 3) 0.4m 4) 0.2m

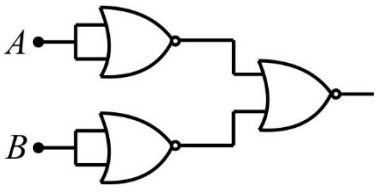
8. A ray of light travelling in a medium of refractive index μ is incident at an angle θ on a composite transparent plate consisting of 50 plates of refractive indices $1.01\mu, 1.02\mu, 1.03\mu, \ldots\ldots 1.50\mu$.

The ray emerges from the composite plate into a medium of refractive index 1.6μ at angle x. Then

1) $\sin x = \left(\frac{1.01}{1.5}\right)^{50} \sin\theta$ 2) $\sin x = \frac{5}{8}\sin\theta$

3) $\sin x = \frac{8}{5}\sin\theta$ 4) $\sin x = \left(\frac{1.5}{1.01}\right)^{50} \sin\theta$

9. The combination of the gates shown in the figure produces:



1) NOR gate 2) OR gate 3) AND gate 4) XOR gate

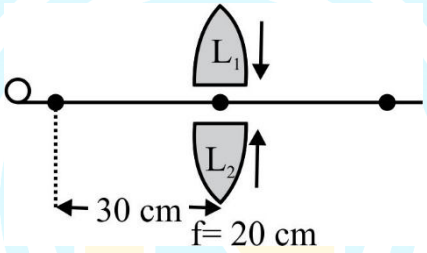
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10. The relation between U, p and V for an ideal gas in an adiabatic process is given by relation

$U = a + bpV$. Find the value of adiabatic exponent (γ) of this gas:

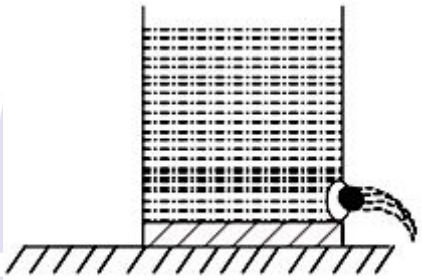
1) $\frac{b+1}{b}$ 2) $\frac{b+1}{a}$ 3) $\frac{a+1}{b}$ 4) $\frac{a}{a+b}$

11. A point object O is placed at a distance of 0.3 m from a convex lens (focal length 0.2 m) cut into two halves and each displaced by 0.1cm each in fig. Find the distance between the images.



1) 0.003 m 2) 0.002 m 3) 0.004 m 4) 0.001 m

12. Liquid of density ρ is filled in the container of negligible mass having area A. Area of hole 'a' is very small (a << A). Co-efficient of friction between container and surface is equal to $\left(\frac{a}{A}\right)$. As liquid is coming out of vessel, instantaneous acceleration of the container.



1) $\left(\frac{A}{a}g\right)$ 2) $\frac{3g}{2}$ 3) $\left(\frac{ag}{A}\right)$ 4) zero

13. A thin spherical conducting shell of radius R has a charge q. Another charge Q is placed at the centre of the shell. Calculate electrostatic potential at a point P a distance R/2 from the centre of the shell.

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1) $\frac{2Q}{2\pi\varepsilon_0 R}-\frac{2q}{4\pi\varepsilon_0 R}$

2) $\frac{Q}{2\pi\varepsilon_0 R}+\frac{q}{4\pi\varepsilon_0 R}$

3) $\frac{(q+Q)}{2\pi\varepsilon_0}\frac{2}{R}$

4) $\frac{2Q}{2\pi\varepsilon_0 R}$

14. Air column in an open organ pipe of length $\ell = 30\,cm$ resonates with a sonic source emitting wave of frequency $f = 1100 H_z$. If the pressure amplitude of the standing wave is $0.1\,P_0$, where P_0 is the atmospheric pressure. Find the difference of the maximum and minimum pressure in the tube at a distance 5 cm from the open end. Take speed of sound in air equal to 330 m/s and ignore the end correction.

1) $\frac{\sqrt{3}}{20}P_0$ 2) $\frac{\sqrt{3}}{10}P_0$ 3) $\sqrt{3}P_0$ 4) $2P_0$

15. **Assertion A:** The Bohr model is applicable to hydrogen and hydrogen- like atoms only.

Reason R: The formulation of Bohr model does not include repulsive force between electrons.

In the light of the above statements, choose the ***correct*** answer from the options given below

1) **A** is true but **R** is false

2) Both **A** and **R** are true and **R** is the correct explanation of **A**

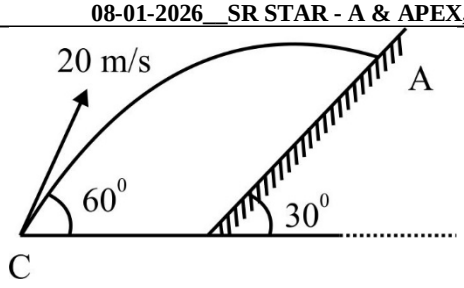
3) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**

4) **A** is false but **R** is true

16. A ball is projected from the point O with velocity 20 m/s at an angle of 60° with horizontal as shown in figure. At highest point of its trajectory it strikes a smooth plane of inclination 30° at point A. The collision is perfectly inelastic. The maximum height from the ground attained by the ball is $\frac{75}{k}$ meter.

Find the value of k? $\left(g = 10\,m/s^2\right)$

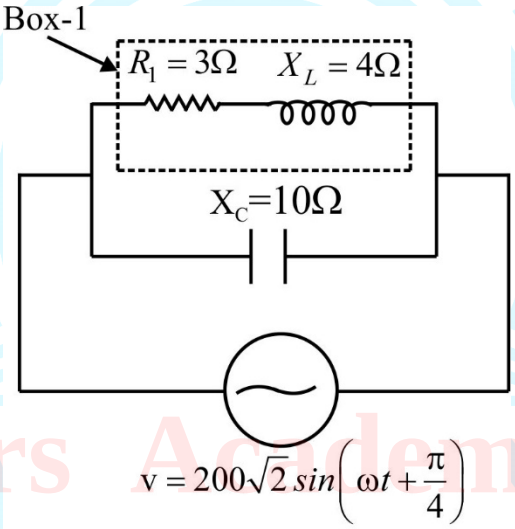
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1) 3 2) 4 3) 5 4) None of these

17. As shown in figure, Instantaneous current in branch having capacitor C will be:

(given $\tan^{-1}(4/3) = 53^0$)



1) $20\sqrt{2}\sin(\omega t + 3\pi/4)$ 2) $40\sqrt{2}\sin(\omega t + \pi/4)$

3) $60\sqrt{2}\sin(\omega t - \pi/4)$ 4) $40\sqrt{2}\sin(\omega t + \pi/4 - 53^0)$

18. Match the **LIST – I** with **LIST – II**

	LIST – I		**LIST – II**
A	Boltzmann constant	I	ML^2T^{-1}
B	Coefficient of viscosity	II	$ML\,T^{-3}K^{-1}$
C	Planck's constant	III	$ML^2T^{-2}K^{-1}$
D	Thermal conductivity	IV	$ML^{-1}T^{-1}$

Choose the ***correct*** answer from the options given below:

1) $A-III, B-II, C-I, D-IV$ 2) $A-II, B-III, C-IV, D-I$

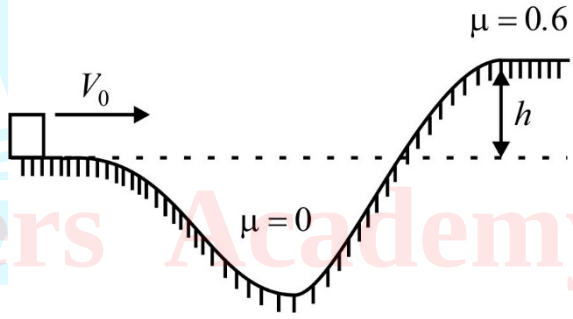
3) $A-III, B-IV, C-II, D-I$ 4) $A-III, B-IV, C-I, D-II$

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19. Certain amount of an ideal gas is contained in a closed vessel. The vessel is moving with a constant velocity v. the rise in temperature of the gas when the vessel is suddenly stopped is (M is molecular mass, $\gamma = \frac{C_p}{C_V}$)

1) $\frac{Mv^2(\gamma-1)}{2R}$ 2) $\frac{Mv^2(\gamma+1)}{2R}$ 3) $\frac{Mv^2}{2R_y}$ 4) $\frac{Mv^2}{2R(\gamma+1)}$

20. In the figure, a block slides along a track from one level to a higher level, by moving through an intermediate valley. The track is frictionless until the block reaches the higher level. There a frictional force stops the block in a distance d. The block's initial speed V_0 is $6m/s$, the height difference h is $1.1m$ and the coefficient of kinetic fraction μ is 0.6. The value of d is $(g = 10\,m/s^2)$



1) $1.17m$ 2) $1.71m$ 3) $3.41m$ 4) $2.81m$

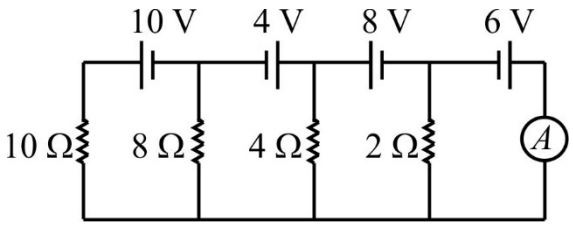
SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. If the ionization potential of hydrogen atom is13.6eV, the energy required to remove electron from the third orbit of hydrogen atom is k/2 eV. Find the value of k…..

22. If the reading of the ideal ammeter connected in the given circuit is $n\left(\frac{13}{20}\right)$ ampere, find the value of n. Assume that the cells have negligible internal resistance.

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23. A beam of light converges at a point P. A converging lens of focal length 25 cm is placed in the path of the beam before converging it at P at a distance of 20 cm from the point P. How far from the lens does the beam converge now?

24. The excess pressure inside a soap bubble A in air is half the excess inside another soap bubble B in air. If the volume of the bubble A is n times the volume of the bubble B, then, the value of n is____

25. A physical quantity C is related to four other quantities p, q, r and s as follows $C = \frac{pq^2}{r^3\sqrt{s}}$ The percentage errors in the measurement of p, q, r and s are 1%, 2%, 3% and 2% , respectively. The percentage error in the measurement of C will be_____%.

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. The K_p of a reaction is 10 atm at 300 K and 4 atm at 400K. The incorrect statement about the reaction following $A(g) \rightleftharpoons B(g) + C(g)$ is

1) The forward reaction is exothermic

2) The E_a of forward reaction is more than that of backward reaction

3) At equilibrium the change in any of the thermodynamic property $(\Delta H, \Delta S$ and $\Delta G)$ is zero

4) The difference between heat of reaction at constant pressure and that at constant volume is RT.

27. Current efficiency is defined as the extent of a desired electrochemical reaction divided by the theoretical extent of the reaction times 100%. What is the current efficiency of an electrodeposition of Cu metal in which 15.0 g Cu is deposited from $aq.Cu^{2+}$ solution by passage of 3A current for 270 minutes.

[Atomic weight of Cu = 63.5, 1.0F=96500C]

1) 0.9 % 2) 49.9 % 3) 93.8 % 4) 51.1%

28. Identify the correct statement/s:

a) Work and heat are the only forms of energy that are exchanged between system and surroundings during a process.

b) Entropy change of surroundings takes place only due to heat exchange between system and surroundings.

c) Increase of entropy of system associated with decrease of enthalpy of system always the process is spontaneous.

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d) In isothermal reversible process, the entropy change of system is directly proportional to heat exchanged between system and surroundings.

1) a, b, c only 2) a, b, d only 3) a, c, d only 4) a, b, c, d

29. Statement I : 1^0 alcohol (RCH_2OH) is oxidized to carboxylic acid $(RCOOH)$ with alkaline $KMnO_4$ and with pyridinium dichromate (PDC) it is oxidized to an aldehyde $(RCHO)$

Statement II : In the reduction of MnO_4^- / OH^- to MnO_2 number of electrons involved per mole are 3 and in the reduction of $Cr_2O_7^{2-} / H^+$ to Cr^{3+} the number of electrons involved per mole are 6.

1) Statement I is correct and Statement II is incorrect

2) Statement I is incorrect and Statement II is correct

3) Both Statement I and II are correct

4) Both Statement I and II are incorrect

30. Mass of Alluminum required to produce 22.4 L of hydrogen gas at STP on reaction with excess of dil. HCl is

Given: Atomic mass of *Al* is 27 $g\ mol^{-1}$

1) 9 g 2) 18 g 3) 7 g 4) 14 g

31. Statement I : In the reaction

$$H_3C-CH_2-CH_2-\underset{\underset{(NMe_3)^+\ OH^-}{|}}{CH}-CH_3 \xrightarrow{\text{heat}} P\,(\text{major product})$$

the product (P) is pent-2-ene

Statement II: More substituted alkene is more stable

1) Both Statement I and II are correct

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2) Both Statement I and II are incorrect

3) Statement I is correct and Statement II is incorrect

4) Statement I is incorrect and Statement II is correct

32. Statement I : Aliphatic carboxylic acids up to nine carbon atoms are colourless gases at room temperature

Statement II : Carboxylic acids are having higher boiling point than their corresponding aldehyde.

1) Statement I is incorrect and Statement II is correct

2) Statement I is correct and Statement II is incorrect

3) Both Statement I and II are incorrect

4) Both Statement I and II are correct

33. The correct order of matching of complex compound in column I with the properties in column II:

	COLUMN I		COLUMN II
A	$\left[Cr\left(NH_3\right)_6\right]^{3+}$	P	Tetrahedral and paramagnetic
B	$\left[Co\left(CN\right)_6\right]^{3-}$	Q	Octahedral and diamagnetic
C	$\left[Ni\left(CN\right)_4\right]^{2-}$	R	Octahedral and paramagnetic
D	$\left[Ni\left(Cl\right)_4\right]^{2-}$	S	Square planar and diamagnetic

1) A-R; B-Q; C-P; D-S

2) A-Q; B-R; C-P; D-S;

3) A-R; B-Q; C-S; D-P

4) A-Q; B-R; C-S; D-P

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34. Choose the correct combination

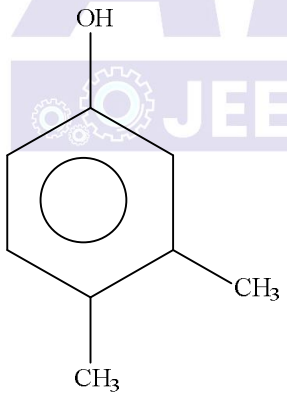
S.No	Vitamin	Source	Deficiency disease
a)	A	Fish liver oil & Carrots	Xeropthalmia
b)	B_{12}	Meat & Fish	Pernicious Anaemia
c)	C	Citrus fruits & Green leafs vegetables	Scruvey
d)	D	Fish, Egg yolk & Sunlight	Rickets

1) a and b only
2) c and d only
3) a, b and d only
4) All are correct

35. Match the List –I with List – II

	List –I		**List – II**
A	Group -11	i)	Nh
B	Group -12	ii)	Fl
C	Group -13	iii)	Rg
D	Group -14	iv)	Cn

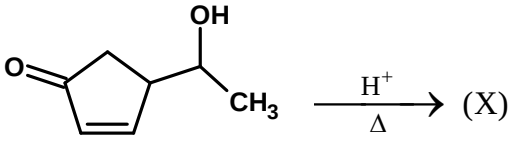
1) $A-ii, B-iv, C-i, D-iii$
2) $A-ii, B-ii, C-i, D-iv$
3) $A-iii, B-iv, C-i, D-ii$
4) $A-ii, B-i, C-iii, D-iv$

36. 

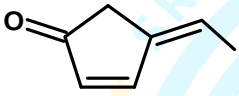
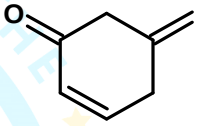
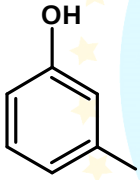
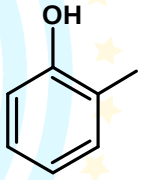
the IUPAC name of the given structure is___

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1) 3, 4 dimethyl phenol
2) 2 methyl 4 hydroxy toluene
3) 2 methyl 5 hydroxy toluene
4) 3 hydroxy xylene

37. 

The major product (X) is

1) 
2) 
3) 
4) 

38. In Dumas' method for estimation of nitrogen 0.8 g of an organic compound gave 120 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound is (Given: Aqueous tension at 300 K = 15 mm Hg)

1) 17.46 %
2) 7.85 %
3) 20.95 %
4) 15.71 %

39. For electrons in '3d' and '4f' orbitals, the orbital angular momentum values, respectively are:

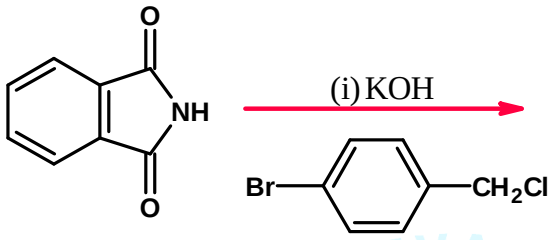
1) $\frac{h}{2\pi}\sqrt{6}\ \ \frac{h}{2\pi}\sqrt{15}$
2) $0,\ \frac{h}{2\pi}\sqrt{2}$
3) $\frac{h}{2\pi}\sqrt{2}\ \ \frac{h}{2\pi}\sqrt{6}$
4) $\frac{h}{2\pi}\sqrt{6}\ \ \frac{h}{2\pi}\sqrt{12}$

40. Which of the following species exhibits the diamagnetic behavior?

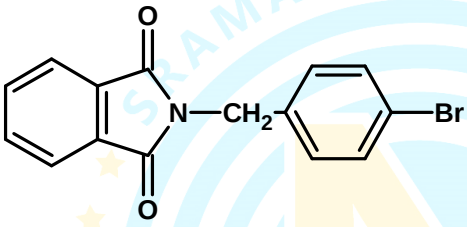
1) NO
2) CO_2
3) O_2^-
4) O_2

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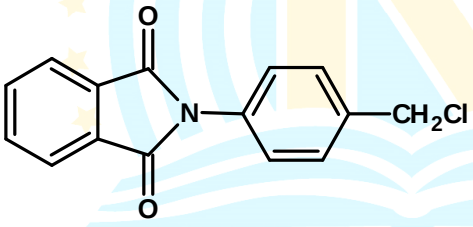
41. The major product of the following reaction is



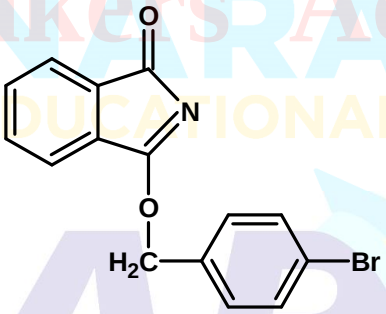
1)



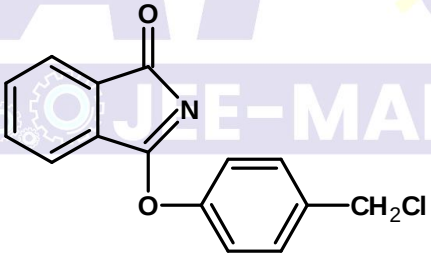
2)



3)

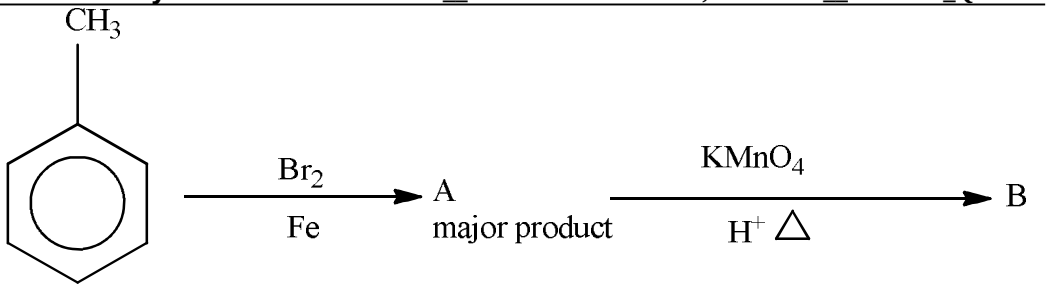


4)



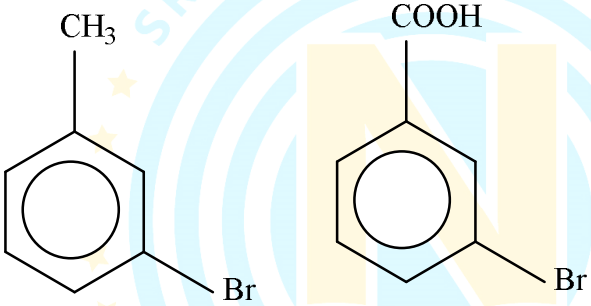
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42.

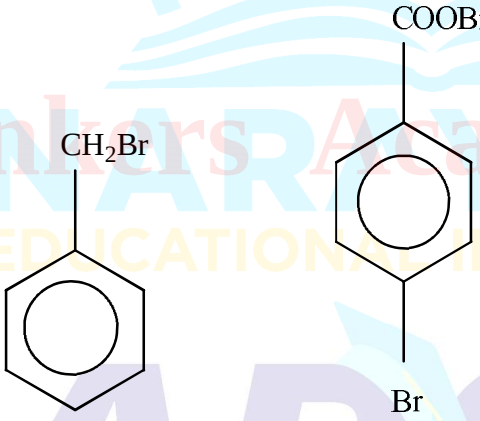


A and B are respectively?

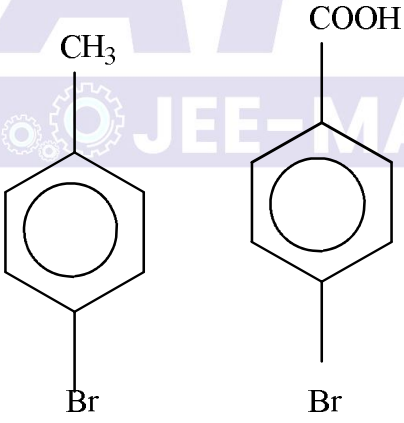
1)



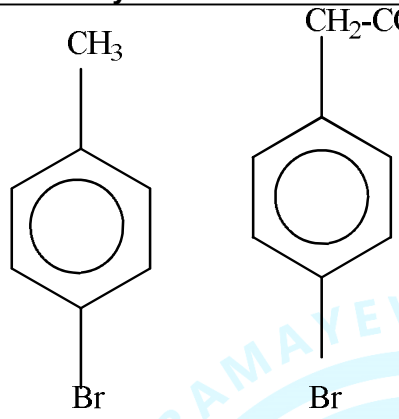
2)



3)



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4) 

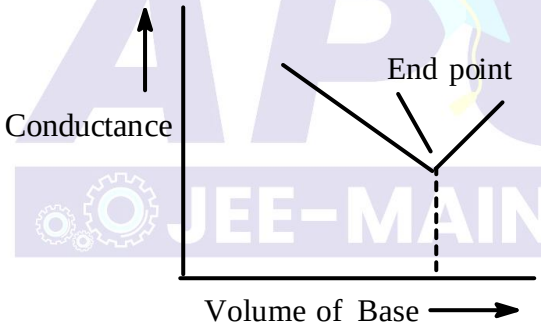
43. 100mL of 0.3$MNaOH$ solution is added to 50 mL of 0.6 M HCl solution kept in a beaker. Now, 15 mL of this mixture is poured into a volumetric flask of 100 mL containing 2 moles of HCl and made the volume upto the mark with distilled water. The solution in this flask is:

1) 35 M HCl solution
2) 25 M HCl solution
3) 0.3 M NaCl solution
4) 0.03 M NaCl solution

44. Which of the following mixture is not an ideal solution / does not obey the Roult's Law

1) Benzene + Toluene
2) n – hexane + n – heptane
3) methanol + ethanol
4) chloroform + Acetone

45. The given graph is for the variation in conductance during titration of 20ml, $0.01M$ H_2SO_4 against standard $NaOH$. Volume of $NaOH$ consumed at equivalence point is 40ml.



Consider the following statements

a) the initial decrease in conductance is due to decrease in $\left[H^+\right]$

b) the molarity of given $NaOH$ solution is $0.01M$

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c) Increase in conductance after equivalence point is due to increase in $\left[SO_4^{2-}\right]$

The correct statements is / are

1) a, b and c
2) b and c only
3) a and b only
4) c only

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

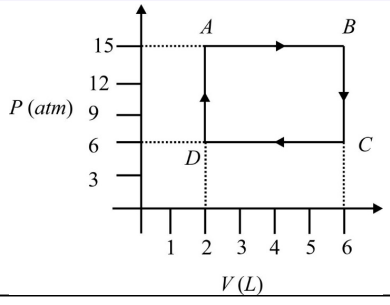
46. The total number of structural isomers possible for the substituted benzene derivatives with the molecular formula C_8H_{10} is_____

47. Among Sc, Mn, Co, Cu, & Zn identify the element with lowest enthalpy of atomization. The spin only magnetic moment value of that element in its +2 oxidation state is____ BM (in nearest integer)

48. X g of nitrobenzene on sulphonation gave 8.25 g of m – nitrobenzene sulphonic acid. X = _____g . (nearest integer)
[Given : molar mass (in $g\ mol^{-1}$) $C:12, H:1, S:32, N:14$ S:32]

49. A sample of n-heptane (1 g) was completely burn t in excess of oxygen in a bomb calorimeter, whose heat capacity is $2.5\ kJ\ K^{-1}$. As a result of combustion reaction, the temperature of the calorimeter is increased by 4 K. The magnitude of the heat of combustion of heptane at constant volume is_____ kJ mol^{-1} (nearest integer).

50. 1.0 mole of an ideal gas subjected to a cyclic process ABCD as shown in the graph below. The net work done in complete cycle is calculated as $-X\ J$. the value of $'X'$ is____ $(1L-atm=101.3J)$.



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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let $f:R\to R$ be a function defined by $f(x)=\left|\,|x+2|-3|x-1|\,\right|$. If k is the number of points of local maxima of $f(x)$ and L is the number of points of local minima of $f(x)$, then $3k+4L$ is

(1) 14 (2) 11 (3) 7 (4) 13

52. Each of the angles β and γ that a given line makes with the positive y and z axes respectively is half the angle that the line makes with positive x-axis. If β_1,β_2 and γ_1,γ_2 are respectively different values of β and γ obtained then the value of $|\beta_1-\beta_2|+|\gamma_1-\gamma_2|$ is

(1) $\frac{3\pi}{4}$ (2) π (3) $\frac{2\pi}{3}$ (4) $\frac{\pi}{2}$

53. Four distinct points $A(0,1)\ B(3,0)\ C(1,4)$ *and* $D(k,k+1)$ are concyclic, then k^2+1 is

(1) 1 (2) 10 (3)5 (4) 26

54. Let the mean of five observations $x_i, i=1,2,3,4,5$ be 4 and variance is 6 where the observations are $x_1=1$, $x_2=2$, $x_3=a$, $x_4=4$ and $x_5=b$. If $a>b$, then the variance of the set $\{n+x_n\}$ for $n=1,2,3,4,5$ is equal to

(1) 6 (2) 12 (3) 9 (4) 15

55. Consider the family of lines

$x(2\sin\theta+3\cos\theta)+y(3\sin\theta-2\cos\theta)-(\cos\theta+5\sin\theta)=0$ where $\theta\in R$.

If P is a point through which these lines pass and the distance of P from the point Q $(2,5)$ is d, then the value of d^2 is

(1) 29 (2) 25 (3) 15 (4) 17

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56. Let $A=\{-4,-3,-2,-1,0,1,2,3\}$. Let R be a relation on A defined by xRy iff $y=\min\{x,1\}$. Let P be the number of elements in R. Let q and r be the minimum number of elements required to be added in R to make it reflexive and symmetric relations, respectively. Then the value of $2p+q+3r$ is equal to

(1) 24 (2) 12 (3) 18 (4) 15

57. If the quadratic equation $x^2-4(k-1)x+(3k-2)=0$ has equal positive roots, then the distance of $p(k,k+1)$ from the line $4x-3y+16=0$ is equal to

(1) $\frac{11}{4}$ (2) 10 (3) 3 (4) 2

58. Let L_1 is the line $3x-y=0$ and L_2 is the line $x-3y=0$, which intersect at origin O. In the first quadrant $p_1, p_2, p_3, p_4, \ldots\ldots p_{11}$ are 11 points on the line L_1 and $q_1, q_2, q_3, \ldots\ldots q_{10}$ are 10 points on the line L_2. Then the number of triangle that can be formed having vertex at three of the 22 points $O, p_1, p_2, \ldots\ldots p_{11}, q_1, q_2 \ldots\ldots q_{10}$ is equal to

(1) 1045 (2) 660 (3) 1155 (4) 605

59. Let $I=\int_0^{\pi}\frac{2xdx}{8\sin^2 x+\cos^2 x}$, then the value of $6\sqrt{2}$ I is____

(1) $6\pi^2$ (2) $\frac{\pi^2}{3}$ (3) $\frac{3\pi^2}{2}$ (4) $3\pi^2$

60. Let f be a function such that $f(x)+5f\left(\frac{36}{x}\right)=3x, x\neq 0$ then the value of $f(2)+f(6)+f(18)$ is equal to

(1) 13 (2) 19 (3) 17 (4) 23

61. The area of the region $\{(x,y):|2x-y|\le y\le 2\sqrt{x}\}$ is

(1) $\frac{8}{3}$ sq units (2) $\frac{32}{3}$ sq units (3) $\frac{40}{3}$ sq units (4) $\frac{35}{3}$ sq units

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62. If the domain of the function $f(x)=\log_3\left(1+\log_{0.5}\left(x^2-5x+6\right)\right)$ is $(\alpha,\beta)\cup(\gamma,\delta)$ then $\alpha+\beta+\gamma+\delta$ is equal to

(1) 18 (2) 12 (3) 14 (4) 10

63. If the probability that the random variable X takes the value x is given by $P(X=x)=\frac{\lambda(3x+4)}{2^x}$ $x=0,1,2,3.....$ where λ is a constant, then $P(x\geq 2)$ is equal to

(1) $\frac{15}{28}$ (2) $\frac{13}{28}$ (3) $\frac{17}{28}$ (4) $\frac{23}{28}$

64. Let $y=y(x)$ be the solution of the differential equation $\frac{dy}{dx}-2\left(\cot^2 x\right)y=2y+\operatorname{cosec}^2 x$ and $y\left(\frac{\pi}{2}\right)=-\frac{1}{2}+e^2$, then $8y\left(\frac{\pi}{4}\right)$ is equal to

(1) 2 (2) $-\frac{1}{2}+e^{-4}$ (3) $-\frac{1}{2}+e^4$ (4) 4

65. If $Z_1,Z_2,Z_3\in C$ are the vertices of an equilateral triangle, whose centroid is $Z_0=1+i$, then $\sum_{k=1}^{3}\left(2Z_k-Z_0\right)^2$ is equal to

(1) 0 (2) 6i (3) 3i (4) 1

66. The number of solutions of the equation $2\left(1+\sqrt{3}\right)\sin x+\sqrt{3}\cos^2 x=\frac{13}{4-\sqrt{3}}, x\in\left[-3\pi,\frac{7\pi}{2}\right]$ is

(1) 5 (2) 4 (3) 0 (4) 6

67. Let C be a circle of minimum area enclosing the ellipse $E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with eccentricity $\frac{1}{2}$ and foci $(\pm 4,0)$. Let PQR be a variable triangle, whose vertex P is on the circle C and side QR of length 4b is parallel to the major axis of E and contains the point of intersection of E with the negative y-axis. Then the maximum area of the triangle PQR is :

(1) $8\left(2+\sqrt{3}\right)$ (2) $8\sqrt{3}\left(2+\sqrt{3}\right)$

(3) $16\sqrt{3}\left(2+\sqrt{3}\right)$ (4) $32\sqrt{3}\left(2+\sqrt{3}\right)$

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68. The shortest distance between the curves $20y = x^2$ and $x^2 + y^2 - 30x + 224 = 0$ is

(1) $5\sqrt{2} - 1$ (2) $5\sqrt{3} - 1$ (3) $3\sqrt{2} - 1$ (4) $4\sqrt{2} - 1$

69. Distance of the point A $(7,4,9)$ to a line passing through $(1,2,3)$ and whose direction ratios are $2,1,2$ Measured parallel to the line $\frac{x+5}{3} = \frac{y+1}{2} = \frac{2z-7}{6}$ is 'd' then the value of d^2 is equal to

(1) 76 (2) 88 (3) 44 (4) 66

70. The sum $1 + \frac{1+2}{2!} + \frac{1+2+2^2}{3!} + \frac{1+2+2^2+2^3}{4!} \ldots\ldots$ up to infinite terms, is equal to

(1) 2e (2) $e + e^2$ (3) $2e^2 - e$ (4) $e^2 - e$

SECTION – II
(Numerical Value Answer Type)

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71. Let I be the identify matrix of order 3×3 and for the matrix $A = \begin{pmatrix} 3 & 4 & P \\ 2 & 5 & -1 \\ 3 & 6 & 2 \end{pmatrix}$, $|A| = -1$. Let B be the inverse of the matrix $Adj\left(A^{2025}\left(Adj\left(A^{2026}\right)\right)\right)$. Then the absolute value of $|PB + 2I|$ is equal to

72. Let $\left(1 + x + x^2\right)^{12} = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_{24} x^{24}$.

If $a_1 + a_3 + a_5 + \ldots.a_{23} - 91a_2 = 182k$ then k is equal to

73. The equation of hyperbola with foci $(2,4)$ and $(8,4)$ is

$8x^2 - y^2 - \alpha x + \beta y + \gamma = 0$, then the value of $2\alpha + 3\beta - \gamma$ is equal to

74. $\lim\limits_{x \to 0} \left(\frac{\tan^{-1} x}{x}\right)^{\frac{1}{\sin^2 x}} = k$, then the value of $48\left|\log_e k\right|$ is

75. Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$, $\vec{c} = 2\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{d}$ be a vector such that $\vec{b} \times \vec{d} = \vec{c} \times \vec{d}$ and $\vec{a}.\vec{d} = 9$, then $\left|\vec{a} \times \vec{d}\right|^2$ is

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